

Vanik Makaryan

Monitoring honeypot traffic with Suricata



VOCAB!

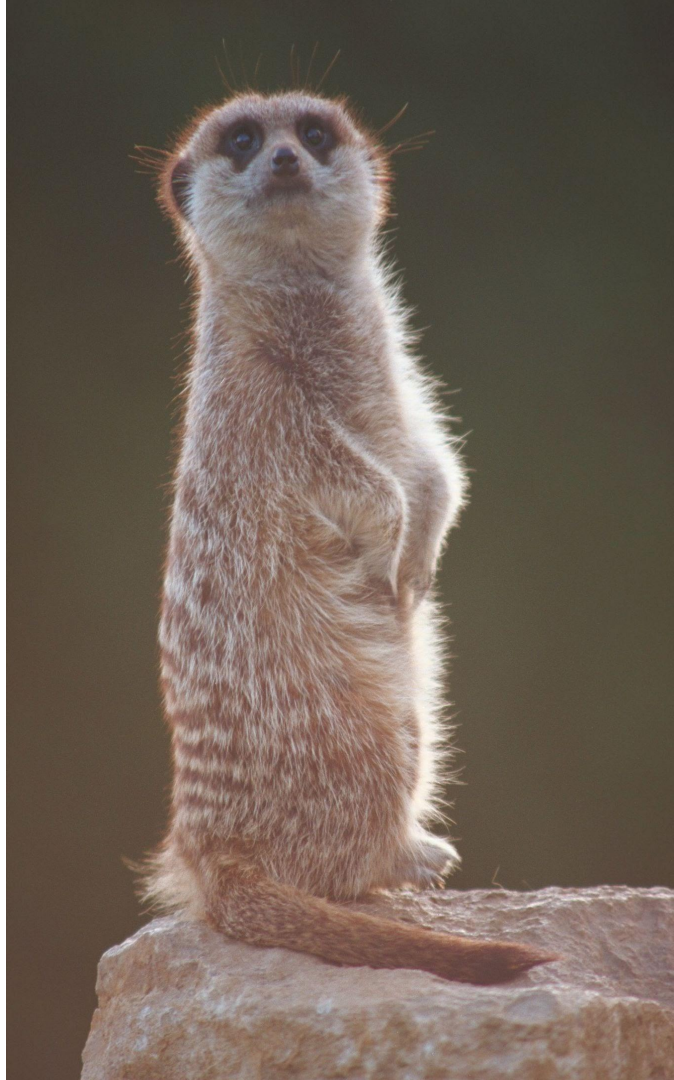
What is a honeypot?

“In computer security terms, a cyber honeypot [is], baiting a trap for hackers. It's a sacrificial computer system that's intended to attract cyberattacks, like a decoy. It mimics a target for hackers, and uses their intrusion attempts to gain information about cybercriminals and the way they are operating or to distract them from other targets.” - kaspersky

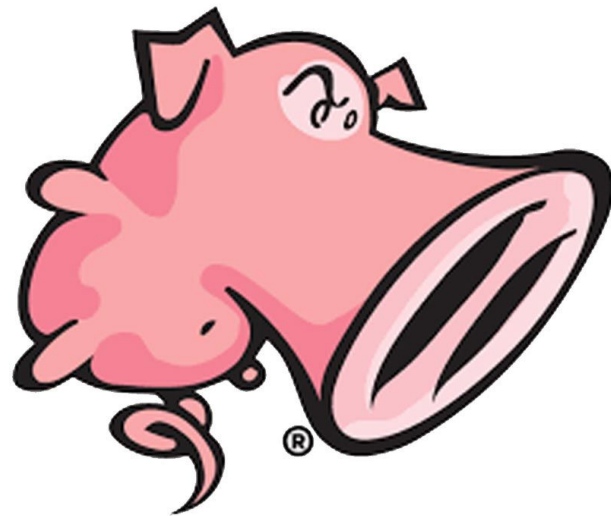


What is an IDS?

An IDS, or Intrusion Detection System, is a way for blue teamers to monitor inbound and outbound traffic for a given network, port, and protocol range (and much more) and generate alerts on potential malicious activity



Meerkat *Suricata suricatta*



What is an incident management system?

An incident management system is a way for companies to organize and link alerts on various systems to one central management platform with the ability to create detailed reports on the alerts called cases.



TheHive



IRIS

servicenow^{*}



What is OSINT?

OSINT stands for open source intelligence. It is essentially the skill for finding information online.





Tools Time



Suricata

Suricata is an open-source IDS.

It is not only an IDS but can be configured to be used as an IPS, or Intrusion Prevention System.



SURICATA®



Alert Rule Anatomy

```
alert http any any -> $HOME_NET 8080 (msg:"ET EXPLOIT Linksys Router Unauthenticated Remote Code Execution"; flow:to_server,established; http.method; content:"POST"; http.uri; content:".cgi"; nocase; http.header; content:"Authorization|3a 20|Basic"; http.request_body; content:"%74%74%63%70%5f%69%70%3d%2d%68%20%60"; fast_pattern; reference:url,sans.org/reading-room/whitepapers/malicious/analyzing-backdoor-bot-mips-platform-35902; classtype:attempted-user; sid:2022758; rev:4; metadata:created_at 2016_04_25, updated_at 2020_10_06;)
```

- alert http any any -> \$HOME_NET 8080
- http.uri; content:".cgi"
- http.header; content:"Authorization|3a 20|Basic";
- http.request_body; content:"%74%74%63%70%5f%69%70%3d%2d%68%20%60"; fast_pattern;



Anatomy of a Gummy Bear

SANS ISC DShield Honeypot

The DShield Honeypot is a honeypot developed by the SANS Internet Storm Center, a research division in the SANS institute.

This honeypot includes some tasty services that may lure attackers such as: HTTP servers and ssh/telnet servers.



TheHive

TheHive is an open source incident management system.

According to their website, StrangeBee, the makers of TheHive, has services like a cloud version of TheHive that is trusted by companies like Cisco and BMW!

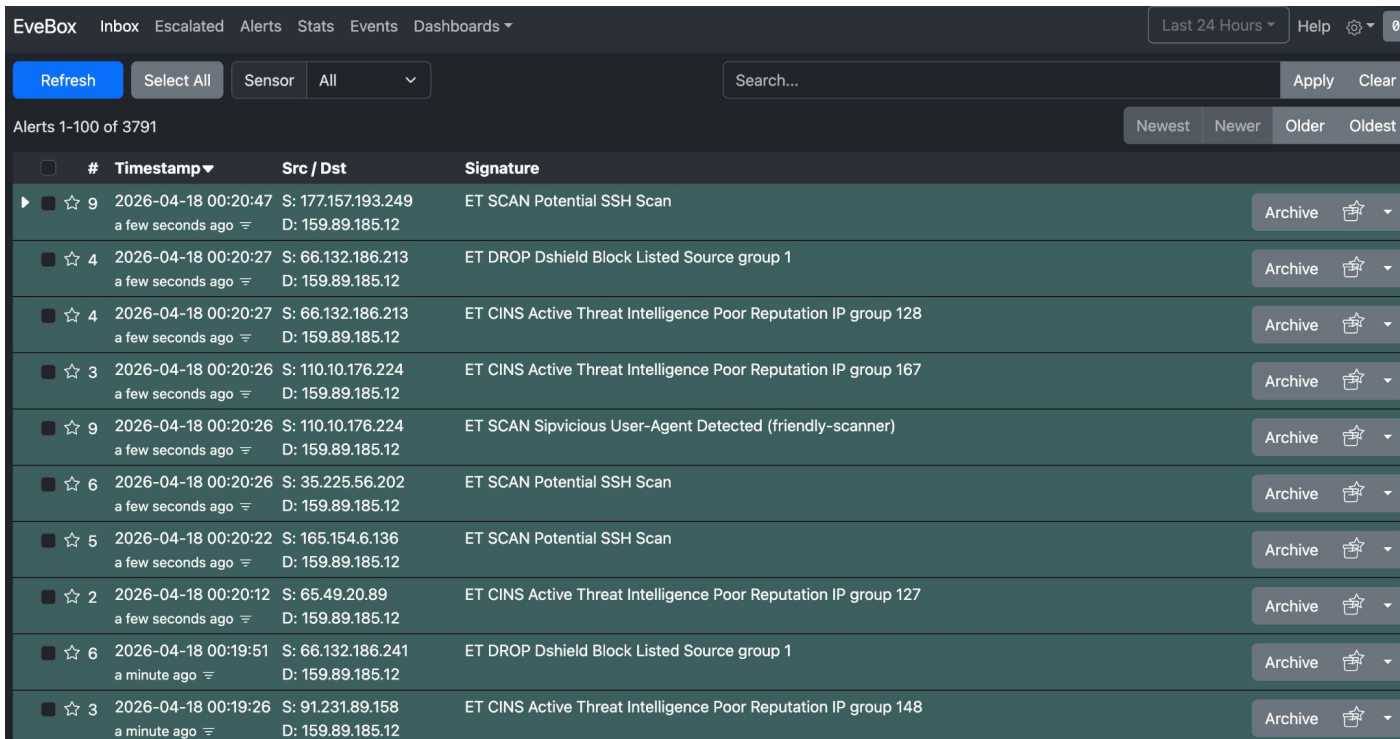


TheHive

Evebox

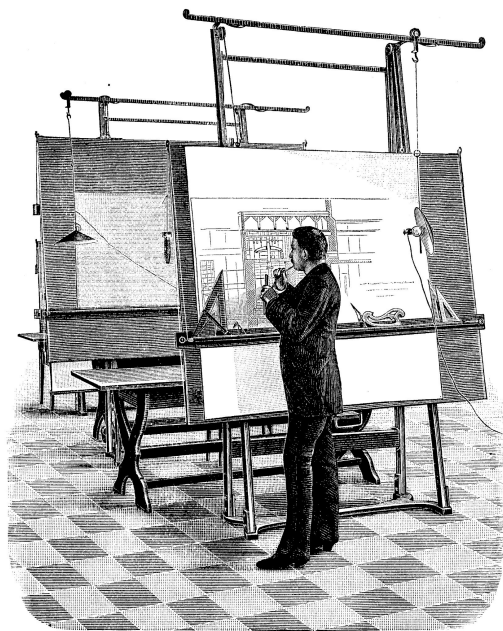
Evebox is pretty simple, just a UI for eve.json

Very lightweight which is always a plus



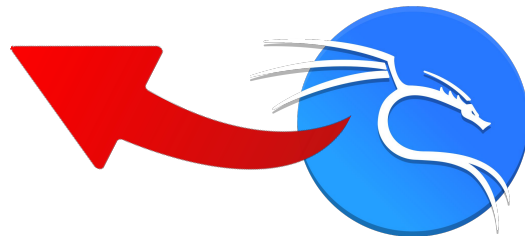
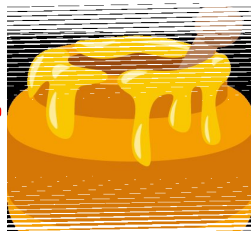
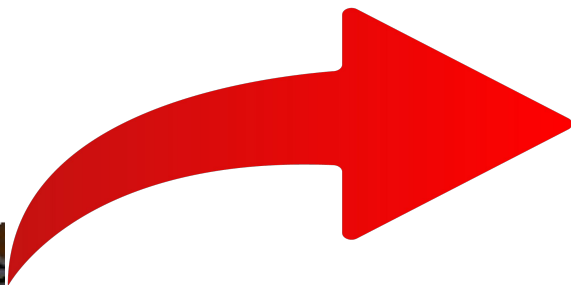
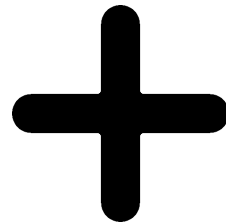
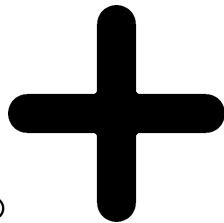
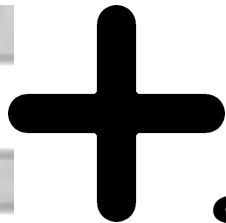
The screenshot shows the EveBox web interface. At the top, there's a navigation bar with links: EveBox, Inbox, Escalated, Alerts, Stats, Events, and Dashboards. On the right of the navigation bar, there's a 'Last 24 Hours' filter, a 'Help' link, and a user profile icon. Below the navigation bar, there's a toolbar with a 'Refresh' button, a 'Select All' button, a 'Sensor' dropdown menu, and an 'All' dropdown menu. To the right of the toolbar is a search bar with the placeholder text 'Search...'. Below the toolbar, there's a header for the alerts list: 'Alerts 1-100 of 3791'. To the right of this header are four buttons: 'Newest', 'Newer', 'Older', and 'Oldest'. The main content area is a table with the following columns: #, Timestamp, Src / Dst, and Signature. Each row represents an alert. The first row has a play button icon, a star icon, and the number 9. The second row has a star icon and the number 4. The third row has a star icon and the number 4. The fourth row has a star icon and the number 3. The fifth row has a star icon and the number 9. The sixth row has a star icon and the number 6. The seventh row has a star icon and the number 5. The eighth row has a star icon and the number 2. The ninth row has a star icon and the number 6. The tenth row has a star icon and the number 3. Each row also has an 'Archive' button and a dropdown menu icon. The table is styled with a dark background and light text.

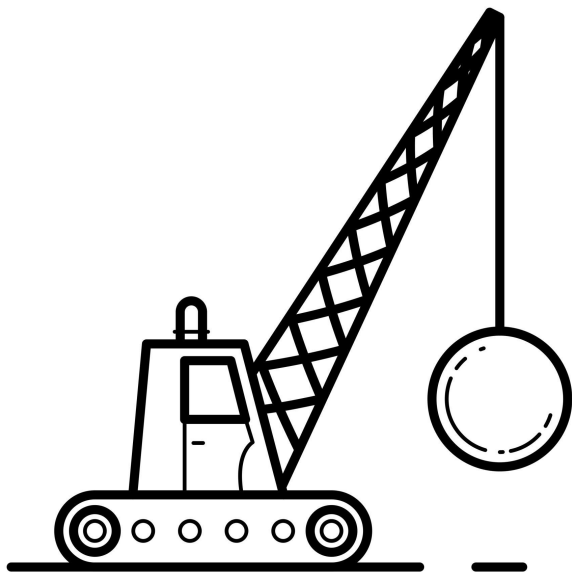
#	Timestamp	Src / Dst	Signature
9	2026-04-18 00:20:47 a few seconds ago	S: 177.157.193.249 D: 159.89.185.12	ET SCAN Potential SSH Scan
4	2026-04-18 00:20:27 a few seconds ago	S: 66.132.186.213 D: 159.89.185.12	ET DROP Dshield Block Listed Source group 1
4	2026-04-18 00:20:27 a few seconds ago	S: 66.132.186.213 D: 159.89.185.12	ET CINS Active Threat Intelligence Poor Reputation IP group 128
3	2026-04-18 00:20:26 a few seconds ago	S: 110.10.176.224 D: 159.89.185.12	ET CINS Active Threat Intelligence Poor Reputation IP group 167
9	2026-04-18 00:20:26 a few seconds ago	S: 110.10.176.224 D: 159.89.185.12	ET SCAN Sipvicious User-Agent Detected (friendly-scanner)
6	2026-04-18 00:20:26 a few seconds ago	S: 35.225.56.202 D: 159.89.185.12	ET SCAN Potential SSH Scan
5	2026-04-18 00:20:22 a few seconds ago	S: 165.154.6.136 D: 159.89.185.12	ET SCAN Potential SSH Scan
2	2026-04-18 00:20:12 a few seconds ago	S: 65.49.20.89 D: 159.89.185.12	ET CINS Active Threat Intelligence Poor Reputation IP group 127
6	2026-04-18 00:19:51 a minute ago	S: 66.132.186.241 D: 159.89.185.12	ET DROP Dshield Block Listed Source group 1
3	2026-04-18 00:19:26 a minute ago	S: 91.231.89.158 D: 159.89.185.12	ET CINS Active Threat Intelligence Poor Reputation IP group 148



Homelab Architecture



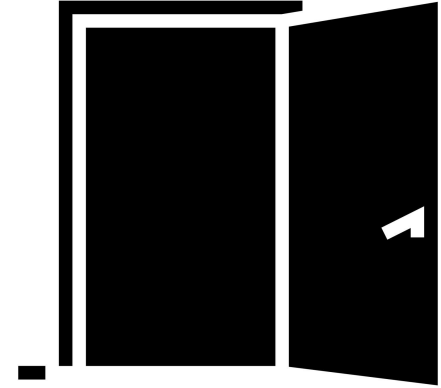




Demo Time!

Lets take a look at an alert!

This is a realtek vulnerability that allows Command execution/backdoors on realtek systems.



ALERT: ET EXPLOIT Realtek SDK - Command Execution/Backdoor Access Inbound (CVE-2021-35394)

Lets figure out what the attacker was trying to do together!

ALERT: ET EXPLOIT Realtek SDK - Command Execution/Backdoor Access Inbound (CVE-2021-35394)

```
alert udp $EXTERNAL_NET any -> $HOME_NET 9034 (msg:"ET EXPLOIT Realtek SDK -  
Command Execution/Backdoor Access Inbound (CVE-2021-35394)"; content:"|00 00 00 00 00  
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00|"; depth:20; content:"|3b|"; distance:0;  
pcre:"/^[^\\x00]*?\\x3b/S"; reference:cve,2021-35394;  
reference:url,blogs.juniper.net/en-us/threat-research/realtek-cve-2021-35394-exploited-in-the  
-wild; classtype:attempted-admin; sid:2033604; rev:1; metadata:attack_target IoT,  
deployment Perimeter, signature_severity Major, created_at 2021_08_17, updated_at  
2021_08_17;)
```

- alert udp \$EXTERNAL_NET any -> \$HOME_NET 9034
- content:"|00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00|"; depth:20;
- content:"|3b|"; distance:0;
- pcre:"/^[^\\x00]*?\\x3b/S";
- example legitimate udp realtek sdk command: set fwup_check_enable 0

Lets conduct some OSINT!

Vote on an alert to look at or first person to say one they find interesting, and we will do a live OSINT hunt together to try and figure out what vulnerability the attacker is trying to exploit, reports on their IP, and maybe even information about the machine they are running!



Generate your own alerts!

Look in the SSL discord channel and you should see a command that involves GET etc/passwd copy that, paste it into your terminal, and your alert should appear!



Homelab design tips

don't be afraid to restart

be flexible in your design - be on the lookout for better ways to do things!

build something that interests you!

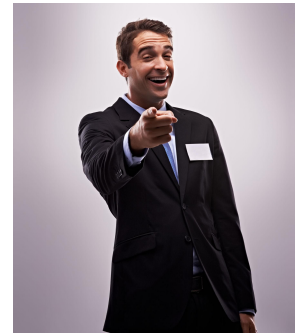


If you liked my presentation.....

I encourage you to build your own!

If you sign up for the Github Student Developer pack you get \$200 in DigitalOcean credits, and as a student, Azure gives you \$100 in Azure credits which is how I was able to do all of this for free!

If you liked my homelab and want to recreate it check out T-Pot or use just use DShield for a simpler setup like I have!



I WANT YOU



To build a homelab

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Any Questions?



Thank you for attending my first ever technical presentation!

